

Effect of the Mediterranean diet intervention on fasting glucose: a dietary intervention study on fasting glucose

Abstract

Background: A dietary intervention study on fasting glucose. Methods: The primary analysis compared the mean fasting glucose (mg/dL) at follow-up between the mediterranean and usual groups using an unpaired Student t-test assuming equal variances, two-sided. Results: see the Results section. Conclusion: the findings are discussed in the context of the analysis sample.

Introduction

We report a dietary intervention study on fasting glucose. The primary question was whether the Mediterranean diet intervention is associated with fasting glucose (mg/dL) at follow-up. The dataset accompanying this report (data.csv) contains one row per participant and is described in the data dictionary.

Methods

Participants and data. Data were collected for 350 participants. Each record contains the variables listed in the data dictionary. Group membership is recorded in the column `diet`.

Exclusion criteria and preprocessing. Only participants aged 18 to 80 years inclusive were included in the analysis. Participants with missing values for glucose_fu were excluded (complete-case analysis). After the exclusions above, observations whose fasting glucose (mg/dL) at follow-up lay more than three sample standard deviations from the overall mean were treated as outliers and removed.

Statistical analysis. The primary analysis compared the mean fasting glucose (mg/dL) at follow-up between the mediterranean and usual groups using an unpaired Student t-test assuming equal variances, two-sided. Group means and sample standard deviations (n - 1 denominator) are reported. The mean difference is reported as mediterranean minus usual. Cohen's d was computed as the mean difference divided by the pooled standard deviation. All analyses were performed on the analysis sample defined above. Statistics were computed with standard scientific software; no random resampling was used.

Results

The primary analysis included 151 participants in the usual group and 161 in the mediterranean group. Mean fasting glucose (mg/dL) at follow-up was 102.70 in the usual group and 98.30 (SD 13.06) in the mediterranean group, a difference of -4.40 ($t = -2.97$, $p = 0.003$; Cohen's $d = -0.336$).

Discussion

The analysis followed the pre-specified plan described in the Methods. Limitations include the observational nature of some comparisons and the exclusion of records with missing values.